MVCUN: A New CNN-Based Autoencoder With Minimum Volume Constraint for Hyperspectral Unmixing

Linqing Wang, Fan Li, Jia Chen, *Member, IEEE*, and Jun Li, *Fellow, IEEE*

***Abstract*—Hyperspectral unmixing (HU) is a fundamental task in hyperspectral image analysis, aiming to decompose mixed pixels into endmember spectra and their corresponding fractional abundances. With the development of deep learning (DL), numerous DL-based unmixing methods have been proposed. However, most of these methods fail to integrate geometric priors (e.g., minimum volume constraints), limiting their performance in nonpure pixel scenarios (i.e., when no pure pixels are present in the scene). Current minimum volume constraint strategies based on the distance between endmembers and the data centroid may yield biased estimations under nonuniform data distributions. To overcome these limitations, we propose a new blind unmixing network with a minimum volume constraint, named MVCUN. By jointly constraining the simplex volume through a total variation (TV) term related to the endmembers and a volume upper bound term, MVCUN achieves improved unmixing performance in nonpure pixel scenarios and exhibits robustness against nonuniform data distributions. Furthermore, we design a multiscale global context attention (MSGCA) module that fuses local and global channel information through multiscale convolutions and fully connected layers, thereby improving the network's ability to estimate abundances. The experimental results on one simulated dataset and three real datasets demonstrate that our MVCUN significantly outperforms existing approaches in terms of both endmember extraction and abundance estimation accuracy.**

***Index Terms*—Autoencoder, convolutional neural network, deep learning (DL), hyperspectral imagery (HSI), hyperspectral unmixing (HU), minimum volume constraint.**

I. Introduction

HYPERSPECTRAL imagery (HSI) is a kind of data acquired through hyperspectral imaging sensors mounted on satellites, aircraft, or uncrewed aerial vehicle (UAV) platforms [1]. These sensors capture detailed spectral response characteristics of surface materials across hundreds of continuous and narrow spectral bands, thereby providing finer spectral resolution compared to conventional imagery [2]. In contrast to regular images, HSIs can significantly improve the accuracy of ground object classification and effectively identify different material compositions. Consequently, hyperspectral remote sensing imagery plays crucial roles in various fields, including environmental monitoring [3], precision agriculture [4], [5], military reconnaissance [6], and mineral exploration [7]. Through meticulous analysis of the spectral information from each pixel, the detailed composition of surface objects can be obtained, thereby providing substantial support for scientific research and decision-making processes.

A. Problem of Mixed Pixels

Due to the physical constraints of sensors, HSIs often exhibit low spatial resolution (typically ranging from meters to hundreds of meters) and high spectral resolution [8]. This results in the fact that individual pixels typically contain spectral information from multiple ground objects, making each pixel's spectral signature a mixture of various endmember (ground material) spectra [9]. The mechanism of mixed pixel formation can be attributed to the complex spatial intermingling of ground objects, where the sensor's spatial resolution proves insufficient to distinguish them accurately, leading to spectral superposition of different materials within each pixel. Specifically, the relative positioning, morphology, and mutual interactions of ground objects in the imaging scene prevent sensors from capturing pure spectral signatures of individual materials, thereby causing spectral mixing. Such a mixed pixel effect substantially increases the difficulty of quantitative analysis and interpreting hyperspectral remote sensing imagery, consequently hindering its practical applications.

B. Hyperspectral Unmixing

To address the mixed pixel effect, hyperspectral unmixing (HU) has emerged as an effective solution. The objective of HU is to separate pure endmembers from mixed spectral signatures and to accurately estimate the abundance of each endmember (i.e., the proportion of each endmember within a pixel). HU techniques can effectively distinguish material compositions and enhance information extraction from the HSI. The unmixing process comprises two fundamental steps: 1) endmember extraction/estimation and 2) abundance estimation (inversion of measured spectra), with approaches that simultaneously estimate both endmembers and abundances being referred to as blind unmixing methods. Endmember

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Linqing Wang and Jun Li are with the School of Computer Science, China University of Geosciences (Wuhan), Wuhan 430079, China.

Fan Li is with Hubei Key Laboratory of Intelligent Geo-Information Processing, School of Computer Science, China University of Geosciences (Wuhan), Wuhan 430074, China, and also with Jiangxi Province Key Laboratory of Smart Water Conservancy, Jiangxi University of Water Resources and Electric Power, Nanchang 330099, China.

Jia Chen is with the School of Computer Science and Technology and Hubei Province Key Laboratory of Intelligent Information Processing and Real-time Industrial System, Wuhan University of Science and Technology, Wuhan 430065, China (e-mail: chen_jia@wust.edu.cn).

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extraction aims to identify the spectral signatures of individual materials from mixed spectra, while abundance inversion involves determining the proportional coefficients of each endmember in mixed pixels through mathematical models. HU is essential for improving the interpretation of hyperspectral remote sensing imagery.

C. Traditional Unmixing Methods

The formulation of spectral mixing models serves as the theoretical cornerstone for HU. Among these, the linear mixing model (LMM)—which assumes that observed spectra can be expressed as linear combinations of endmember spectra—has become the most widely adopted approach due to its clear physical interpretability and computational efficiency [10]. Traditional linear unmixing methods have evolved significantly. In particular, geometric approaches assume that the HSI data reside within (or on) the surface of a simplex formed by endmembers. Here, endmember extraction equates to identifying the simplex's vertex set. When pure pixels for all endmembers exist in the observed data, endmember localization can be achieved through simplex volume maximization criteria [11], [12], [13], [14]. However, when pure pixels are absent from the observed data (i.e., some endmembers are not represented in the dataset), simplex volume maximization-based endmember extraction methods may underperform. To overcome this limitation, simplex volume minimization techniques have been introduced for endmember estimation in scenarios lacking pure pixels [15], [16], [17]. Methods such as [18], [19], and [20] achieve simplex volume reduction by minimizing the determinant of the endmember matrix. Following endmember determination, abundances are typically derived via least squares solutions to the linear model [14]. For blind unmixing in the absence of pure pixels, enhanced nonnegative matrix factorization (NMF) methods with minimum-volume constraints demonstrate unique advantages [21]. These approaches integrate volume constraints into their objective functions through distinct mechanisms: [22] employs a squared simplex volume term as explicit regularization to directly penalize geometric dispersion, while [23] constructs implicit constraints by leveraging Euclidean distances between endmembers and the data centroid, effectively steering endmembers toward dense regions. Li et al. [24] enhances robustness through an initialization-guided regularization that restricts endmember deviations from prior estimates, and [25] innovatively incorporates a total variation (TV) penalty on endmembers, promoting spatial coherence among endmembers and achieving volume minimization by bringing endmembers closer to one another. However, these traditional methods based on minimum simplex volume constraints often perform poorly when applied to highly mixed data, due to the lack of sufficient pixels on the linear simplex faces [10], [26].

D. Deep-Learning-Based Unmixing Methods

Recent years have witnessed rapid advances in deep learning (DL) techniques, opening new avenues for HU research. Capitalizing on their powerful feature learning capabilities, DL methods can automatically extract complex patterns from massive datasets, providing more efficient tools for HU compared to conventional approaches. These data-driven techniques excel at exploiting the high-dimensional characteristics of HSI data while capturing their inherent nonlinear relationships, thereby significantly enhancing unmixing accuracy [27]. The autoencoder, an unsupervised DL architecture comprising encoder and decoder components, has demonstrated particular promise. The encoder learns low-dimensional embeddings of input data, while the decoder reconstructs the original input from these compressed representations. Through iterative training that minimizes reconstruction error as the loss function, autoencoders effectively distill essential low-dimensional features. This architecture proves particularly suitable for blind unmixing tasks: the learned embeddings naturally correspond to abundance maps, while the decoder weights inherently represent endmember signatures [28]. The method [29] was the first to introduce autoencoders into the HU field. It performs data denoising prior to unmixing by cascading a denoising autoencoder with a nonnegative sparse autoencoder. An untied denoising autoencoder with sparsity (uDAS) [30] incorporates additional denoising constraints into the encoder and the $l_{2,1}$ sparsity constraint into the decoder to jointly optimize denoising and unmixing, thereby reducing the extra error introduced by using a separate denoising autoencoder. Stacked nonnegative sparse autoencoders (SNSA) [31] address HU challenges in datasets with outliers and low signal-to-noise ratios (SNRs) by sequentially executing outlier detection and unmixing tasks. The deep autoencoder network (DAEN) [32] ensures the physical constraints of abundance estimation by replacing the unmixing autoencoder in SNSA with a variational autoencoder.

The hidden layers of the autoencoder in the aforementioned methods are simple linear layers that cannot effectively exploit the spatial information of HSI. Therefore, more advanced networks have been employed as encoders to extract spatial information from the data. In [33], a fully convolutional network-based autoencoder architecture was proposed to effectively leverage the spatial information of HSI. The cycle-consistency unmixing network (CyCU-Net) [34] introduced a self-aware loss function, comprising two spectral reconstruction terms and an abundance reconstruction term, to enhance unmixing performance and stability. The multistage convolutional autoencoder network (MSNet) [35] addressed the limitations of single-stage autoencoder networks in terms of receptive field and stability by employing a multistage progressive unmixing strategy, which expands the receptive field while preserving local details. In [36] and [37], deep image priors were applied to HU. Particularly, [37] incorporated a minimum volume constraint to optimize unmixing performance in nonpure pixel scenarios. Due to their strong capability to capture global features, Transformers have also been utilized for unmixing [38], [39], [40], [41]. A U-shaped transformer network (UST-Net) [39] employs Swin Transformer blocks during downsampling and upsampling stages to extract multiscale hyperspectral image features, generating contextually rich and spatially accurate abundance maps while preserving local detail information.

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E. Motivations of This Work

Most existing DL-based unmixing methods fail to effectively incorporate geometric prior information (e.g., minimum volume constraints), which limits their performance in nonpure pixel scenarios. Although some studies have attempted to construct minimum volume constraints based on the distances between endmembers and the data centroid [37], such strategies may lead to biased endmember predictions when the data distribution is nonuniform [21]. To address this issue, we propose MVCUN, a new blind unmixing network based on the minimum volume constraint. By incorporating endmember TV along with a simplex volume upper bound constraint as a minimum volume regularization term within a DL framework, our method enhances unmixing performance in nonpure pixel scenarios and demonstrates increased robustness against nonuniform data distributions. In addition, by integrating a convolutional network with a multiscale global context attention (MSGCA) module, the proposed network improves the exploitation of spatial and spectral information in HSI, thereby significantly enhancing the quality of abundance estimation.

F. Contributions of This Work

The main innovative contributions of our work can be summarized in the following two aspects.

1) A new blind unmixing network, named MVCUN, is proposed, which introduces the TV of the endmembers as a minimum volume constraint in the loss function. Compared to regularization based on the distance between endmembers and the data centroid, this strategy exhibits stronger robustness when dealing with nonuniform data distributions. Furthermore, a novel regularization term is introduced to explicitly constrain the upper bound of the simplex volume, enhancing the performance of the unmixing network.
2) A new MSGCA module is proposed to enable efficient spectral information extraction from HSI. This module integrates local and global channel information through a combination of multiscale convolutions and fully connected layers to generate adaptive channel attention weights. This mechanism effectively suppresses interference from irrelevant bands, enhancing the quality of abundance estimation.

G. Paper Organization

The rest of this article is organized as follows. Section II introduces the LMM and the autoencoder used for unmixing. Section III details our proposed approach. Section IV presents the experimental results along with their analysis. Section V concludes this article with some remarks and hints at plausible future research lines.

II. Related Work

A. Linear Mixing Model

The LMM posits that the spectral signature of each pixel in a hyperspectral image is a linear combination of the spectra of several pure materials, known as endmembers. More precisely, the model represents each pixel's spectrum as a weighted sum of the endmember spectra, with the weights, referred to as abundances, being nonnegative. Mathematically, this can be expressed as

$$\mathbf{Y} = \mathbf{E}\mathbf{A} + \boldsymbol{\eta} \tag{1}$$

where $\mathbf{Y} = [\mathbf{y}_1, \cdots, \mathbf{y}_N] \in \mathbb{R}^{L\times N}$ represents the HSI data consisting of N pixels, each with L spectral bands, $\mathbf{E} = [\mathbf{e}_1, \cdots, \mathbf{e}_p] \in \mathbb{R}^{L\times p}$ denotes the endmember matrix comprising p endmembers, $\mathbf{A} \in \mathbb{R}^{p\times N}$ is the abundance matrix, and $\boldsymbol{\eta} \in \mathbb{R}^{L\times N}$ accounts for noise. Furthermore, to ensure physically meaningful results, the LMM imposes two constraints on the abundances: the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC), which are given by

$$\mathbf{A}_{ij} \geq 0 \tag{2}$$

$$\sum_{i=1}^{p} \mathbf{A}_{ij} = 1. \tag{3}$$

B. Autoencoders for Unmixing

An autoencoder is an efficient unsupervised learning method and the most widely used network in DL-based unmixing approaches. Typically, an autoencoder consists of an encoder and a decoder.

1) *Encoder:* The encoder is responsible for compressing the input data and extracting low-dimensional features. In unmixing, the extracted low-dimensional features can be regarded as data abundance. To satisfy the ANC and ASC constraints, some methods add a Softmax layer in the final stage of the encoder to enforce these constraints. The mathematical expression is as follows:

$$\hat{\mathbf{A}} = \text{Softmax}\left(\text{Encoder}\left(\mathbf{Y}\right)\right) \tag{4}$$

where $\hat{\mathbf{A}}$ represents the obtained abundance, and the Softmax function is defined as

$$\text{Softmax}\left(\mathbf{z}_i\right) = \frac{e^{\mathbf{z}_i}}{\sum_{j=1}^{K} e^{\mathbf{z}_j}} \tag{5}$$

2) *Decoder:* The decoder is responsible for reconstructing the input from low-dimensional features. In unmixing, the decoder is typically a linear layer. The data reconstruction process can be viewed as a linear combination of abundance and endmembers

$$\hat{\mathbf{Y}} = \text{Decoder}\left(\hat{\mathbf{A}}\right) = \hat{\mathbf{E}}\hat{\mathbf{A}} \tag{6}$$

where $\hat{\mathbf{Y}}$ is the reconstructed hyperspectral image, and $\hat{\mathbf{E}}$ represents the weights of the decoder, which can be regarded as the endmember matrix to be extracted.

III. Methodology

A. Network Architecture

Our MVCUN is an end-to-end convolutional autoencoder designed for blind unmixing, as illustrated in Fig. 1. The architecture comprises a convolutional encoder and a linear decoder. The encoder consists of four sequential convolutional modules and processes a spectral cube with spatial dimensions

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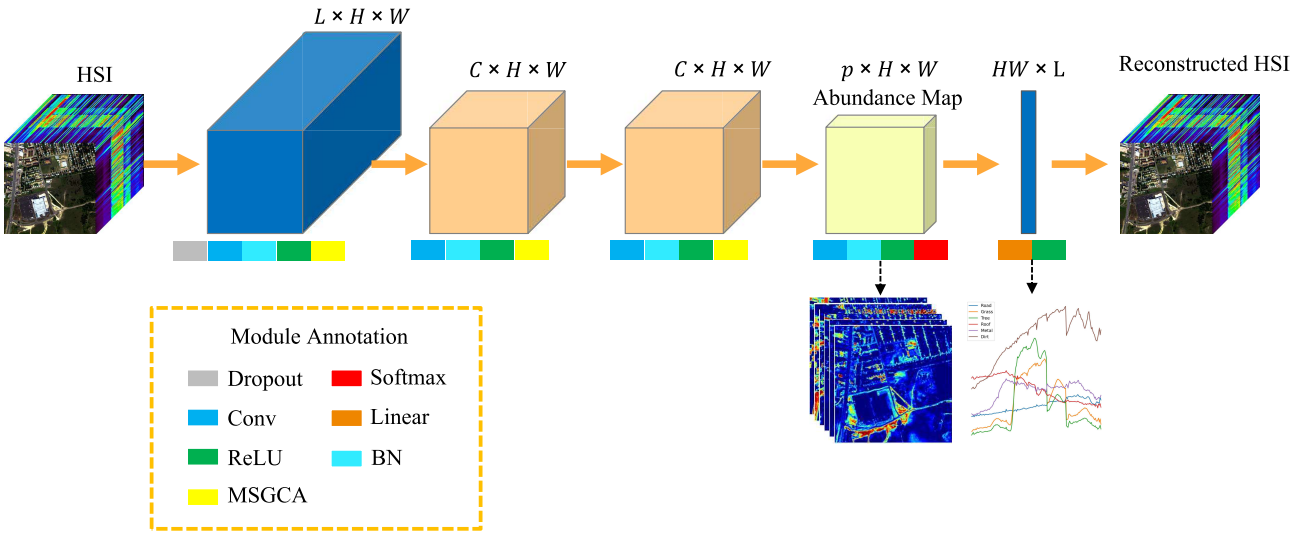


Fig. 1. Architecture of proposed MVCUN. The encoder comprises four convolutional blocks integrated with MSGCA modules for channel attention, while the decoder employs a linear layer. Input-level dropout injects artificial noise, and softmax activation enforces the ANC and ASC.

$H \times W$ and L spectral bands that is used as input to estimate the abundance maps. Subsequently, the linear decoder reconstructs the original input using these estimated abundances, with its weights representing the extracted endmembers.

To enhance resilience against noise, a dropout layer (with a rate of 0.2) is incorporated at the initial encoding stage, introducing a marginal denoising strategy that injects controlled noise perturbations into the input. Drawing inspiration from denoising autoencoders [42], [43], this design leverages stochastic channel masking to train the network to map corrupted inputs to clean spectral signatures. This noise-immune training approach significantly improves feature extraction stability under Gaussian noise, ensuring dependable performance in challenging scenarios.

In the feature encoding stage, each convolutional block (except the last one) follows a strict sequential processing flow: 2-D convolutional layer (Conv2d), batch normalization (BN) layer, rectified linear unit (ReLU) activation function, and MSGCA. The convolutional layers adopt a kernel size of 3×3 with a stride of 1×1. Dynamically calculated padding is applied to preserve the spatial resolution of feature maps. All convolutional layers disable bias terms, relying on BN for feature standardization. The proposed MSGCA mechanism is strategically positioned after nonlinear activation to conduct multiscale channel interactions, optimally recalibrating spectral feature responses across hierarchical layers. The first three convolutional blocks maintain a consistent channel dimension of $C = 128$ to preserve feature richness, while the fourth block performs dimension reduction to match the number of endmembers, p. Finally, a Softmax layer enforces the ANC and ASC, normalizing the endmember abundances within each pixel to unity summation.

In the decoding stage, a fully connected layer linearly maps the low-dimensional abundance vectors back to the original spectral dimension L, and a ReLU activation function guarantees the nonnegativity of the reconstructed spectra.

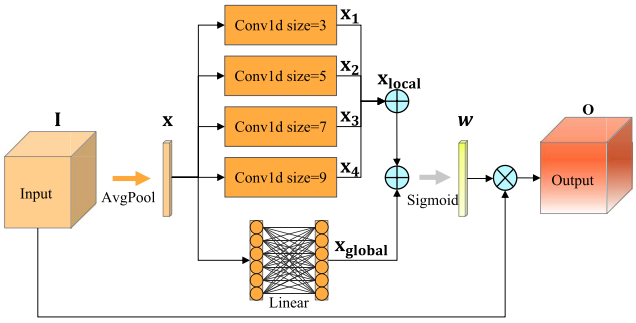


Fig. 2. Structure of the MSGCA module. Multiscale 1-D convolutions extract channel features, while linear layers capture global dependencies. Fused representations generate adaptive channel weights through sigmoid activation.

B. Multiscale Global Context Attention

The MSGCA module is a specially designed channel attention mechanism that achieves dynamic recalibration of spectral channel weights by integrating multiscale local channel features with global channel context. This design effectively enhances the utilization of spectral information in HSI while preserving dimensional integrity. As illustrated in Fig. 2, the MSGCA module employs a dual-branch architecture: one branch captures multiscale local dependencies, while the other extracts global channel features through a dimensionality-preserving fully connected transformation [44].

In the local feature extraction branch, the input feature map $\mathbf{I} \in \mathbb{R}^{C \times H \times W}$ (where C, H, and W denote the number of channels, height, and width, respectively) first undergoes adaptive average pooling to aggregate spatial information, compressing the spatial dimensions while retaining rich spectral characteristics. This yields a channel descriptor $\mathbf{x}$

$$\mathbf{x} = \text{AvgPool}(\mathbf{I}) \in \mathbb{R}^{1 \times C}. \tag{7}$$

The channel descriptor is then processed through parallel multiscale 1-D convolutional branches, enabling the module

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to capture local spectral interactions at multiple scales. Appropriate padding strategies are applied to maintain the original channel dimension. Outputs from different kernel branches are fused via elementwise summation, forming a unified multiscale local feature $\mathbf{x}_{\text{local}}$. The specific procedure is as follows:

$$\mathbf{x}_1 = \text{Conv1d}_1(\mathbf{x}) \tag{8}$$

$$\mathbf{x}_2 = \text{Conv1d}_2(\mathbf{x}) \tag{9}$$

$$\mathbf{x}_3 = \text{Conv1d}_3(\mathbf{x}) \tag{10}$$

$$\mathbf{x}_4 = \text{Conv1d}_4(\mathbf{x}) \tag{11}$$

$$\mathbf{x}_{\text{local}} = \mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3 + \mathbf{x}_4 \tag{12}$$

where $\text{Conv1d}_{1,2,3,4}$ denote 1-D convolutional layers with kernel sizes of 3, 5, 7, and 9, respectively.

Simultaneously, a fully connected layer (Linear) followed by a ReLU activation function is applied to extract the global channel features $\mathbf{x}_{\text{global}}$ without dimensionality reduction, ensuring no loss of dimensional information during feature aggregation. Unlike traditional squeeze-and-excitation mechanisms [45] that employ dimensionality reduction, our design preserves complete spectral characteristics, enabling more effective spectral information modeling. The process is defined as

$$\mathbf{x}_{\text{global}} = \text{ReLU}(\text{Linear}(\mathbf{x})). \tag{13}$$

Finally, the multiscale local features and global contextual information are integrated via additive fusion, followed by a sigmoid activation function σ to generate the final channel attention weights $\mathbf{w}$. These weights are applied to the original input features through channelwise multiplication, achieving adaptive feature recalibration that selectively emphasizes important channels while suppressing irrelevant ones. This is expressed as

$$\mathbf{w} = \sigma\left(\mathbf{x}_{\text{global}} + \mathbf{x}_{\text{local}}\right). \tag{14}$$

This design allows MSGCA to efficiently capture fine-grained spectral relationships while preserving global consistency, making it particularly well-suited for HU tasks.

C. Loss Functions

1) Reconstruct Loss: In the HU task, typical reconstruction loss functions generally comprise the mean squared error (mse) and the spectral angle distance (SAD). MSE effectively controls the discrepancy between the reconstructed images and the original images. However, it is sensitive to spectral scales, which may result in misidentifying the same endmember as different entities when they exhibit similar shapes but different scales due, for instance, to changing illumination conditions [28]. In contrast, SAD—owing to its scale invariance—can circumvent this issue, but it might lead to scale distortions in the extracted endmembers. Therefore, we combine these two metrics to define our reconstruction loss function as follows:

$$\mathcal{L}_{\text{mse}} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{y}_i - \hat{\mathbf{y}}_i)^2 \tag{15}$$

$$\mathcal{L}_{\text{SAD}} = \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\mathbf{y}_i^{\top}\hat{\mathbf{y}}_i}{\|\mathbf{y}_i\|\|\hat{\mathbf{y}}_i\|}\right)\cdot\frac{180}{\pi} \tag{16}$$

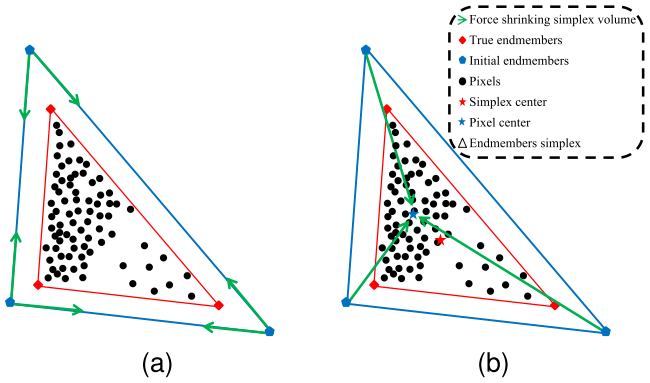


Fig. 3. Schematic of the minimum volume regularization term. (a) TV. (b) Center.

$$\mathcal{L}_{\text{re}} = \mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{SAD}}. \tag{17}$$

2) Minimum Volume Regularization: The endmember TV term reduces the volume of the simplex by pulling the endmembers closer together, as illustrated in Fig. 3(a). This approach is particularly robust when the data points are nonuniformly distributed. Quite opposite, constraints based on the distance between the endmembers and the data center depend on the position of the data center and are susceptible to the influence of skewed data distributions, leading to biased unmixing results, as illustrated in Fig. 3(b). For this reason, we choose to use the endmember TV term as the minimum volume constraint

$$\mathcal{L}_{\text{TV}} = \frac{1}{2}\sum_{i,j=1}^{p}\|\hat{\mathbf{e}}_i - \hat{\mathbf{e}}_j\|_2^2. \tag{18}$$

The endmember TV constrains the volume of the endmember simplex only implicitly. To achieve more precise volume control, we propose a novel constraint term that enforces a stricter constraint by bounding the upper limit of the simplex volume. The volume of a $(p-1)$-dimensional simplex generated $\mathbf{E}$ can be expressed as

$$\text{Vol}(\mathbf{E}) = \frac{1}{(p-1)!}\sqrt{\det(\mathbf{E}'^{\top}\mathbf{E}')} \tag{19}$$

where $\mathbf{E}' = [\mathbf{e}_2 - \mathbf{e}_1, \ldots, \mathbf{e}_p - \mathbf{e}_1] \in \mathbb{R}^{L\times(p-1)}$ contains the centered vectors. For the Gram matrix $G = \mathbf{E}'^{\top}\mathbf{E}'$, the arithmetic mean-geometric mean inequality (AM-GM) implies

$$\det(\mathbf{G}) \le \left(\frac{\text{tr}(\mathbf{G})}{p-1}\right)^{p-1}. \tag{20}$$

Using the identity $\text{tr}(\mathbf{G}) = \|\mathbf{E}'\|_F^2$, the volume upper bound becomes

$$\text{Vol}(\mathbf{E}) \le \frac{1}{(p-1)!}\left(\frac{\|\mathbf{E}'\|_F^2}{p-1}\right)^{(p-1)/2} \propto \|\mathbf{E}'\|_F^2. \tag{21}$$

Therefore, we employ $\|\mathbf{E}'\|_F^2$ to constrain the upper bound of simplex volume. The final loss function for the minimum volume constraint is formulated as follows:

$$\mathcal{L}_{\text{mv}} = \lambda\mathcal{L}_{\text{TV}} + \alpha\|\mathbf{E}'\|_F^2 \tag{22}$$

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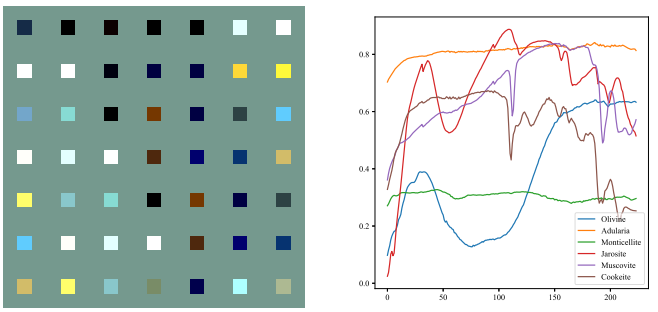


Fig. 4. Simulated dataset. (a) True-color image. (b) Endmembers.

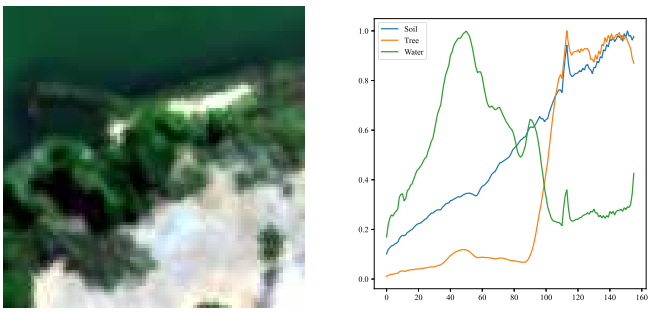


Fig. 5. Samson dataset. (a) True-color image. (b) Endmembers.

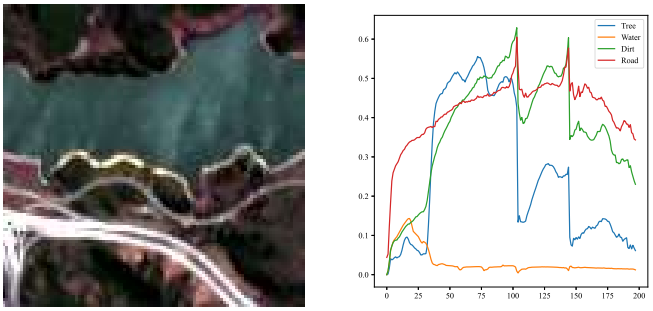


Fig. 6. Jasper Ridge dataset. (a) True-color image. (b) Endmembers.

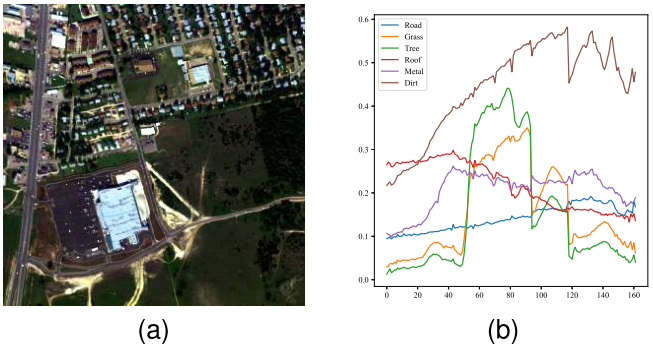


Fig. 7. Urban dataset. (a) True-color image. (b) Endmembers.

where both λ and α are positive regularization coefficients that control the tradeoff between TV penalty and volume constraint.

3) Abundance Sparsity Regularization: Abundance sparsity serves as a critical physical prior in HU, grounded in the spatial aggregation effect of natural terrain distributions (i.e., specific endmembers typically dominate mixing processes within localized regions). To enhance the sparse representation capability of abundance estimation and improve unmixing quality, we employ a sparse constraint compliant with the requirements outlined in [46], formulated as

$$\mathcal{L}_{\text{abu}} = \beta \frac{1}{p} \sum_{i=1}^{p} \frac{\|\hat{\mathbf{A}}_i\|_1}{\sqrt{N}\|\hat{\mathbf{A}}_i\|_2} \tag{23}$$

where β denotes the regularization coefficient. The complete loss function is subsequently defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{re}} + \mathcal{L}_{\text{mv}} + \mathcal{L}_{\text{abu}}. \tag{24}$$

IV. Experiments

A. Datasets

We conducted experiments on one simulated dataset and three real-world datasets. A description of the considered datasets follows.

1) *Simulated dataset:* To validate MVCUN's robustness in nonpure pixel scenarios, we employ the synthetic dataset from [27] as shown in Fig. 4(a). This carefully designed benchmark intentionally violates the pure-pixel assumption, and contains 105 × 105 pixels with 224 spectral bands. The scene incorporates six mineral endmembers [see Fig. 4(b)] curated from the United States Geological Survey (USGS) spectral library: Olivine, Adularia, Monticellite, Jarosite, Muscovite, and Cookeite. These components simulate realistic mixing phenomena through controlled spatial distributions.
2) *Samson dataset:* Acquired using the Spectroscopic Aerial Mapper with On-board Navigation (SAMSON) hyperspectral sensor [47], this benchmark dataset contains 952 × 952 pixels with 156 spectral channels spanning 401–889 nm at 3.13-nm spectral resolution. To ensure computational efficiency, a representative 95 × 95 subimage [see Fig. 5(a)] is typically employed for analysis. The scene comprises three characteristic endmembers—Soil, Tree, and Water—whose spectral signatures are systematically characterized in Fig. 5(b).
3) *Jasper Ridge dataset:* Collected by the Airborne Visible Infra-Red Imaging Spectrometer (AVIRIS) [48], this dataset contains 512 × 614 pixels with 224 spectral bands covering 380–2500 nm (spectral resolution: 9.46 nm). Following standard preprocessing procedures, 26 bands affected by atmospheric absorption (bands 1–3, 108–112, 154–166, and 220–224) were removed, retaining 198 valid bands. As shown in Fig. 6(a), experimental analyses commonly utilize a 100×100 subimage. Spectral unmixing identifies four dominant endmembers in this scene [see Fig. 6(b)]: Road, Soil, Water, and Tree.
4) *Urban dataset:* Captured by the hyperspectral data collection experiment (HYDICE) sensor, this dataset features a complex urban landscape with 307 × 307 pixels at 2-m spatial resolution [see Fig. 7(a)]. The original 210-band data (400–2500-nm spectral range, 10-nm

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TABLE I
HYPERPARAMETERS SETTING

dataset	λ	α	β	lr	epochs
Simulated	0.001	0.001	0.001	0.001	3000
Samson	4	0.001	0.2	0.001	6000
JasperRidge	1.5	0.001	0.2	0.001	6000
Urban	0.6	0.001	2	0.001	1000

resolution) underwent rigorous preprocessing, removing 48 bands (1–4, 76, 87, 101–111, 136–153, and 198–210) affected by noise and atmospheric interferers, resulting in 162 analyzable bands. This dataset is particularly notable for its spectral complexity, containing six characteristic endmembers [see Fig. 7(b)]: Road, Grass, Tree, Roof, Metal, and Dirt. Its accurate representation of material interactions in urban environments establishes it as a critical benchmark for HU validation.

B. Baselines and Metrics

To validate the effectiveness of the proposed method, we selected several popular HU algorithms as baseline methods for comparison. These include traditional linear unmixing method VCA-FCLS [13], [14], CNN-based methods such as CNNAEU [33], MSNet [35], CyCU-Net [34], and MiSiCNet [37], as well as the probabilistic model PGMSU [49], and Transformer-based methods DeepTrans-HSU [38] and UST-Net [39]. Notably, both MiSiCNet and PGMSU incorporate minimum volume constraints, albeit in different forms: MiSiCNet employs a constraint based on the distance between endmembers and the data centroid, whereas PGMSU utilizes a TV regularizer on the endmembers.

To quantitatively evaluate the quality of the unmixing results, we employ two metrics: SAD and the root mean square error (RMSE). The SAD metric measures the spectral similarity between the extracted endmembers $\hat{\mathbf{E}}$ and the reference endmembers $\mathbf{E}$, where a smaller SAD value indicates better endmember extraction accuracy as it reflects smaller spectral deviations. The RMSE metric evaluates the accuracy of the estimated abundance maps, with smaller values corresponding to more precise abundance estimation. For clearer comparison, all RMSE values are scaled by a factor of 100. The mathematical formulations of these metrics are given below

$$\mathrm{SAD}\left(\mathbf{E},\hat{\mathbf{E}}\right)=\frac{1}{p}\sum_{i=1}^{p}\arccos\left(\frac{\mathbf{e}_i^{\top}\hat{\mathbf{e}}_i}{\|\mathbf{e}_i\|\|\hat{\mathbf{e}}_i\|}\right)\cdot\frac{180}{\pi} \tag{25}$$

$$\mathrm{RMSE}\left(\mathbf{A},\hat{\mathbf{A}}\right)=100\times\sqrt{\frac{1}{pn}\sum_{i=1}^{p}\sum_{j=1}^{N}\left(\mathbf{A}_{ij}-\hat{\mathbf{A}}_{ij}\right)^2}. \tag{26}$$

C. Hyperparameters Setup

In the network architecture, we uniformly set the size of all convolutional kernels to three. The first three convolutional layers are configured with 128 output channels, while the final convolutional layer is set to have p output channels to match the number of abundance maps. We employ the Adam optimizer with a fixed learning rate of 0.001, and the specific number of training epochs can be found in Table I.

The regularization coefficients in the loss function are critical parameters that significantly influence the method's performance. For the regularization coefficient λ, we set it to 0.001 for synthetic datasets, while for real datasets, we determine its optimal value through a search within the range of 0.5–5. The parameter α is fixed at 0.001 across all experiments. As for β, it is set to 0.001 for synthetic datasets, whereas for real datasets, we conduct a search within the range of 0.1–3 to identify its optimal value. The detailed optimal parameter configurations are provided in Table I. For other comparative methods, we made appropriate adjustments based on the default parameters provided in their respective references to ensure optimal performance.

In addition, to ensure fairness, all methods requiring endmember initialization were initialized using VCA.

D. Experiments With Simulated Data

To verify the robustness of the proposed method against noise, Gaussian white noise at three different levels (20, 30, and 40 dB) was added to the Simulated dataset, and comparative experiments were conducted with several benchmark algorithms. The quantitative evaluation of the endmember extraction results is shown in Table II. As the noise level increases, the performance of all methods deteriorates; however, our method consistently achieves the best results across the three noise levels with only minor performance degradation, demonstrating its robust endmember extraction capability in the presence of noise.

It is worth noting that, due to the violation of the pure pixel assumption in the simulated dataset, methods incorporating a minimum volume constraint (such as MiSiCNet and PGMSU) perform better than other methods on the endmember SAD metric. Nonetheless, their performance is still inferior to that of our method, which further validates the superiority of the constraint term designed in our approach.

Furthermore, the quantitative evaluation of the abundance estimation results is presented in Table III. Under all noise conditions, our method significantly outperforms the competing algorithms, with a substantial lead over the second-best approach. This clearly demonstrates the effectiveness of the proposed MSGCA module in enhancing the accuracy of abundance estimation.

To further illustrate the overall effectiveness of our approach, we visualized the experimental results obtained at a 20-dB noise level (see Fig. 8). The visualization reveals that both the endmember extraction and the abundance estimation results are in close agreement with the ground truth.

E. Experiments With Real Data

1) Samson: The endmember extraction results are shown in Table IV. Our method achieves the best performance for the Soil endmember and, while its performance for the Tree and Water endmembers is second-best, the overall average SAD is optimal. The quantitative results for abundance estimation are presented in Table V, where the abundance maps for all endmembers demonstrate that our method yields the best

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TABLE II

MEAN SAD COMPARISON ON THE SIMULATED DATASET UNDER DIFFERENT NOISE LEVELS. THE BEST PERFORMANCES ARE SHOWN IN RED, AND THE SUBOPTIMAL PERFORMANCES ARE SHOWN IN BLUE

SNR (dB)	VCA	CNNAEU	PGMSU	MiSiCNet	CyCU-Net	MSNet	DeepTrans-HSU	UST-Net	MVCUN
20	8.39	10.29	**4.60**	7.37	13.83	5.91	6.09	9.63	**1.71**
30	4.89	10.86	5.04	**3.41**	15.08	3.60	5.99	4.87	**1.44**
40	4.23	10.72	4.55	**2.77**	14.41	3.52	6.06	3.13	**0.88**

TABLE III

OVERALL RMSE COMPARISON ON THE SIMULATED DATASET UNDER DIFFERENT NOISE LEVELS. THE BEST PERFORMANCES ARE SHOWN IN RED, AND THE SUBOPTIMAL PERFORMANCES ARE SHOWN IN BLUE

SNR (dB)	FCLS	CNNAEU	PGMSU	MiSiCNet	CyCU-Net	MSNet	DeepTrans-HSU	UST-Net	MVCUN
20	7.66	11.94	6.35	**3.92**	14.97	6.52	6.47	19.28	**2.02**
30	7.45	7.81	6.30	**2.19**	15.23	8.53	5.52	35.57	**1.08**
40	8.36	7.51	6.10	**1.79**	15.43	7.27	5.75	36.01	**0.89**

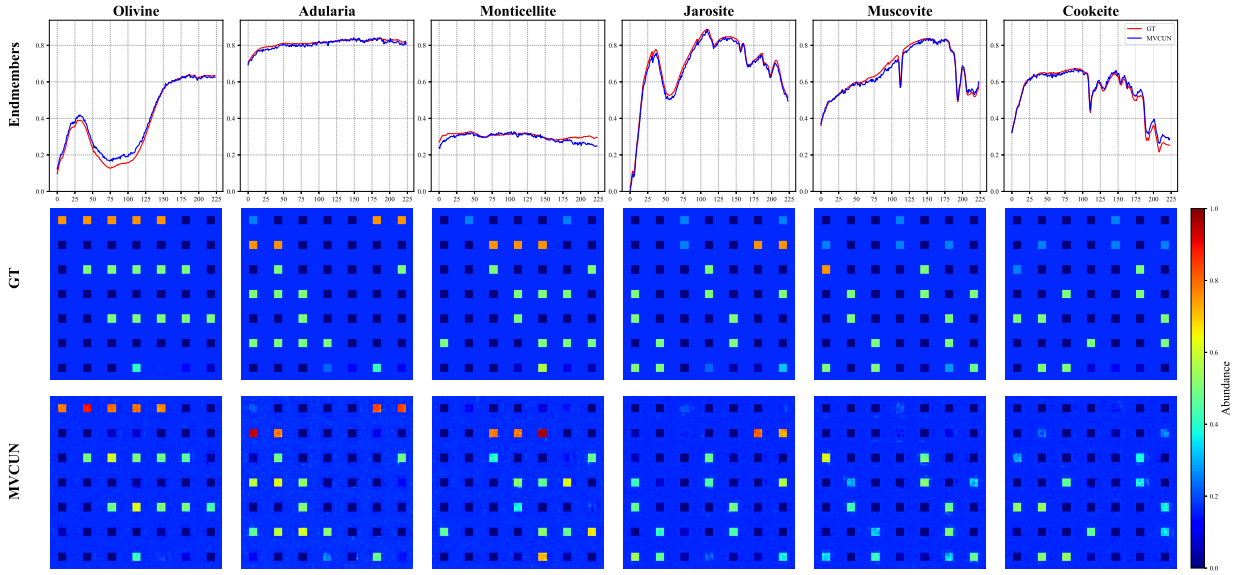


Fig. 8. Endmember extraction and abundance estimation performance on the simulated dataset (SNR: 20 dB).

TABLE IV

SAD COMPARISON ON THE SAMSON DATASET. THE BEST PERFORMANCES ARE SHOWN IN RED, AND THE SUBOPTIMAL PERFORMANCES ARE SHOWN IN BLUE

	VCA	CNNAEU	PGMSU	MiSiCNet	CyCU-Net	MSNet	DeepTrans-HSU	UST-Net	MVCUN
Soil	2.15	0.70	0.68	**0.52**	2.88	0.74	0.71	1.07	**0.49**
Tree	3.68	2.18	2.21	**1.52**	2.18	1.95	3.22	2.13	**1.80**
Water	7.42	2.23	16.46	5.56	9.05	2.79	**1.33**	2.31	**1.90**
Mean	4.42	**1.70**	6.45	2.53	4.70	1.83	1.76	1.84	**1.40**

TABLE V

RMSE COMPARISON ON THE SAMSON DATASET. THE BEST PERFORMANCES ARE SHOWN IN RED, AND THE SUBOPTIMAL PERFORMANCES ARE SHOWN IN BLUE

	FCLS	CNNAEU	PGMSU	MiSiCNet	CyCU-Net	MSNet	DeepTrans-HSU	UST-Net	MVCUN
Soil	22.26	15.05	16.08	15.12	18.41	**5.60**	14.18	7.17	**3.67**
Tree	18.36	10.59	12.19	20.39	19.44	**4.30**	8.37	6.65	**1.75**
Water	33.56	13.20	23.19	27.03	15.60	**4.35**	12.91	6.33	**3.48**
Overall	25.55	13.08	17.75	21.41	17.89	**4.79**	12.08	6.72	**3.09**

results. Furthermore, the visual comparison of the endmember extraction and abundance estimation (shown in Figs. 9 and 10) clearly indicates that the recovered abundance maps from our method are the closest to the ground truth.

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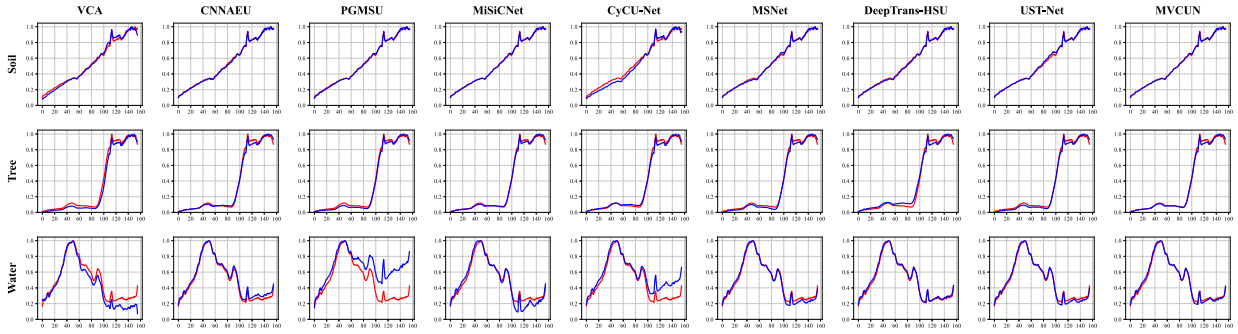


Fig. 9. Visual comparison of extracted endmembers using different unmixing methods and the corresponding ground truth endmembers in the Samson dataset: blue represents extracted endmembers, red represents ground truth endmembers.

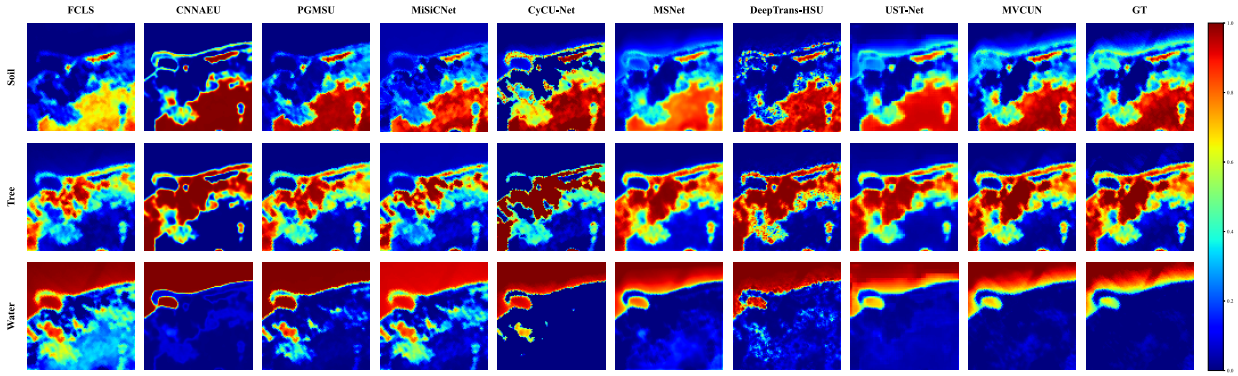


Fig. 10. Visual comparison of abundance maps from the Samson dataset obtained by the different unmixing methods.

TABLE VI

SAD COMPARISON ON THE JASPER RIDGE DATASET. THE BEST PERFORMANCES ARE SHOWN IN RED, AND THE SUBOPTIMAL PERFORMANCES ARE SHOWN IN BLUE

	VCA	CNNAEU	PGMSU	MiSiCNet	CyCU-Net	MSNet	DeepTrans-HSU	UST-Net	MVCUN
Tree	11.06	3.56	3.29	5.80	8.49	**1.18**	2.92	3.99	**1.78**
Water	10.95	2.05	8.10	4.99	5.11	1.72	4.92	**1.69**	**1.65**
Dirt	4.02	6.12	**1.56**	3.65	6.68	6.61	5.90	2.34	**1.43**
Road	9.95	7.41	**1.51**	8.10	5.16	2.73	4.92	2.11	**1.44**
Mean	9.00	4.78	3.62	5.64	6.36	3.06	4.66	**2.53**	**1.57**

TABLE VII

RMSE COMPARISON ON THE JASPER RIDGE DATASET. THE BEST PERFORMANCES ARE SHOWN IN RED, AND THE SUBOPTIMAL PERFORMANCES ARE SHOWN IN BLUE

	FCLS	CNNAEU	PGMSU	MiSiCNet	CyCU-Net	MSNet	DeepTrans-HSU	UST-Net	MVCUN
Tree	21.35	11.88	10.30	10.83	9.94	9.47	7.73	**4.23**	**2.47**
Water	4.43	8.73	11.97	11.53	10.14	3.91	12.04	**3.19**	**1.48**
Dirt	30.97	15.16	**8.28**	14.33	14.40	9.54	13.30	10.28	**2.53**
Road	20.14	11.03	**6.02**	12.52	11.80	8.42	12.20	9.73	**2.88**
Overall	21.45	11.93	9.41	12.37	11.71	8.17	11.51	**7.56**	**2.40**

2) Jasper Ridge: Compared to the Samson dataset, the Jasper Ridge dataset features a greater number of endmembers and more complex land cover distributions; however, our method still achieves excellent performance. Table VI presents the SAD metrics for various methods across different endmembers. As shown in the table, our method achieves the best performance for the Soil, Road, and Water endmembers, and the second-best performance for the Tree endmember. The final average SAD is significantly lower than that of the competing methods, which fully demonstrates the high accuracy and robustness of our approach in terms of endmember extraction. Furthermore, Table VII shows the RMSE metrics for abundance estimation. For the Tree, Soil, Road, and Water endmembers, our method consistently achieves the best performance, with an overall RMSE of only 2.40—substantially lower than that of the other methods. This indicates that our

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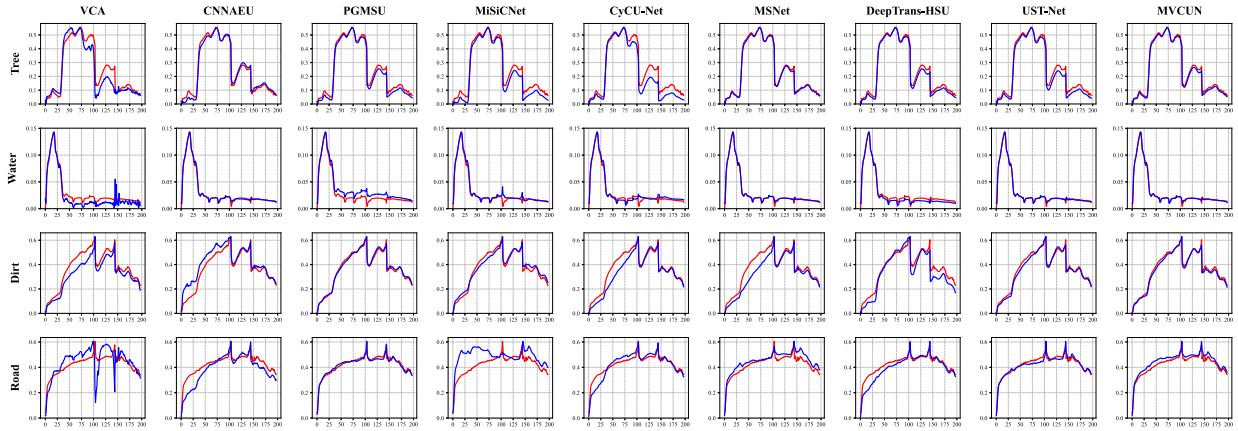


Fig. 11. Visual comparison of extracted endmembers using different unmixing methods and the corresponding ground truth endmembers in the Jasper Ridge dataset: blue represents extracted endmembers, red represents ground truth endmembers.

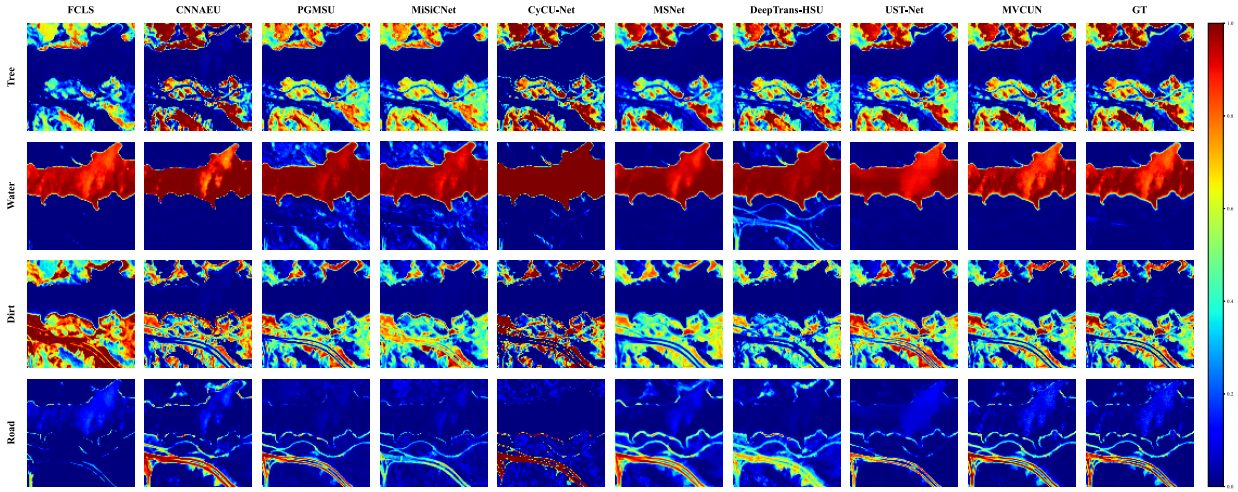


Fig. 12. Visual comparison of abundance maps from the Jasper Ridge dataset obtained by the different unmixing methods.

TABLE VIII

SAD COMPARISON ON THE URBAN DATASET. THE BEST PERFORMANCES ARE SHOWN IN RED, AND THE SUBOPTIMAL PERFORMANCES ARE SHOWN IN BLUE

	VCA	CNNAEU	PGMSU	MiSiCNet	CyCU-Net	MSNet	DeepTrans-HSU	UST-Net	MVCUN
Road	14.98	3.14	4.45	6.34	8.48	**2.86**	3.61	2.95	**1.80**
Grass	11.51	4.35	24.86	11.97	62.13	**3.34**	5.92	**1.93**	8.93
Tree	13.61	10.04	**4.59**	**3.18**	7.03	8.73	10.68	7.18	5.19
Roof	17.46	**2.48**	2.98	5.21	47.75	**2.20**	12.58	18.40	4.68
Metal	35.24	23.50	**7.12**	26.47	26.87	8.34	20.98	18.36	**5.70**
Dirt	25.63	4.84	6.05	**1.80**	19.96	10.05	5.77	13.22	**3.37**
Mean	19.74	8.06	8.34	9.16	28.70	5.92	9.92	10.34	**4.95**

approach offers higher precision in abundance estimation and can more accurately recover the true abundance distributions of the endmembers. In addition, Figs. 11 and 12 provide visual comparisons of the abundance maps and the extracted endmembers, respectively, in the Jasper Ridge dataset. The figures clearly illustrate that the differences between the endmembers extracted by our method and the ground truth endmembers are minimal, and the spatial distributions of the abundance maps are highly consistent with the actual conditions.

3) Urban: The Urban dataset is the most challenging among the three considered real-world datasets, as it contains the highest number of endmembers and a larger spatial size, posing greater challenges for unmixing methods. The quantitative results of endmember extraction are presented in Table VIII, showing that most methods struggle to accurately extract all endmembers. However, our method still achieves the best overall performance. The quantitative results of abundance estimation are shown in Table IX. Our method achieves the best performance on all endmembers except for

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TABLE IX

RMSE COMPARISON ON THE URBAN DATASET. THE BEST PERFORMANCES ARE SHOWN IN RED, AND THE SUBOPTIMAL PERFORMANCES ARE SHOWN IN BLUE

	FCLS	CNNAEU	PGMSU	MiSiCNet	CyCU-Net	MSNet	DeepTrans-HSU	UST-Net	MVCUN
Road	21.35	12.57	25.41	18.83	30.26	**12.20**	21.75	18.70	**11.13**
Grass	44.91	27.93	28.13	22.18	37.88	19.78	35.85	**13.20**	**12.53**
Tree	31.32	11.16	26.97	24.95	27.74	12.29	39.56	**8.21**	**11.14**
Roof	14.72	9.72	19.77	13.89	11.60	9.45	18.36	**7.19**	**6.41**
Metal	18.88	26.37	11.00	35.63	12.28	16.65	13.44	**9.90**	**7.29**
Dirt	36.25	**10.46**	19.34	17.46	23.77	16.07	18.46	14.26	**13.07**
Overall	29.84	18.08	22.55	23.23	25.73	14.81	26.40	**12.55**	**10.57**

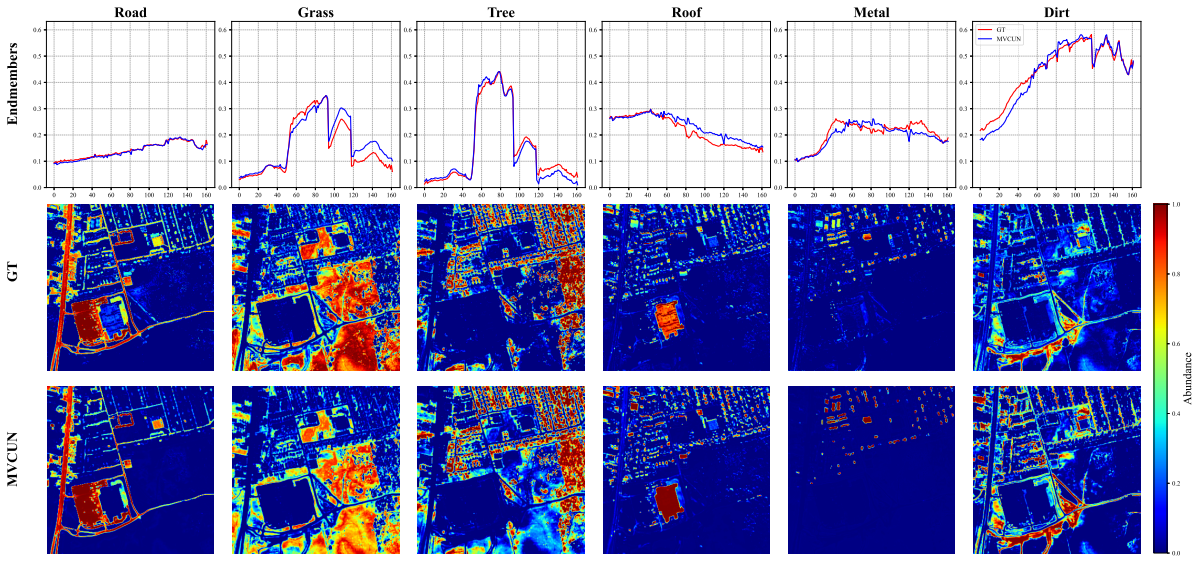


Fig. 13. Endmember extraction and abundance estimation performance on the Urban dataset.

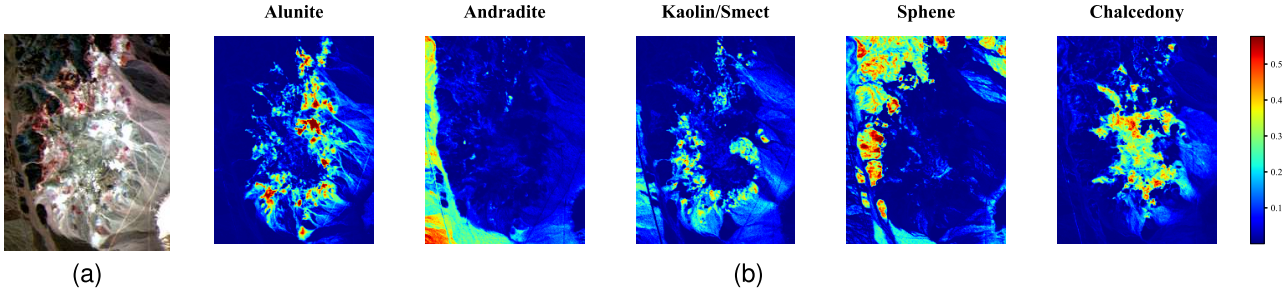


Fig. 14. Cuprite dataset. (a) True-color image. (b) Abundance estimation for the Cuprite dataset.

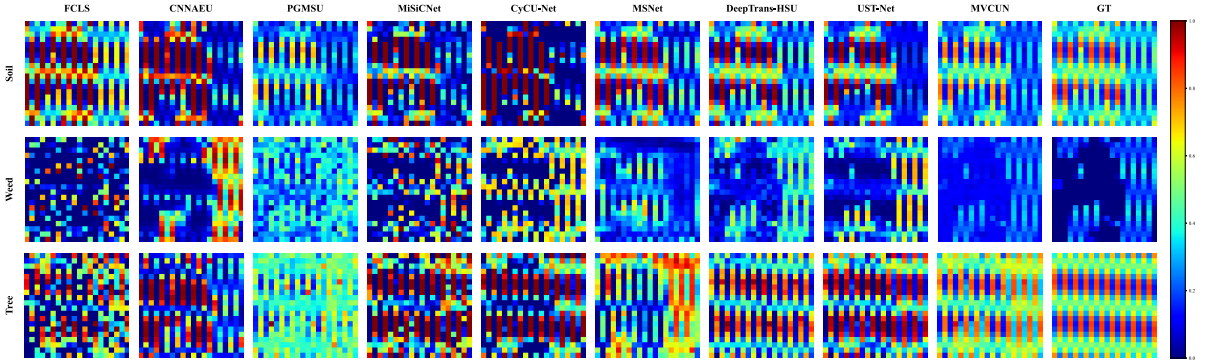


Fig. 15. Visual comparison of abundance maps from the Orchard dataset obtained by the different unmixing methods.

Tree, where it ranks second. The overall RMSE is the lowest among all competing methods, demonstrating its superiority in abundance estimation. In addition, Fig. 13 illustrates the visualized results of the abundance maps and endmember

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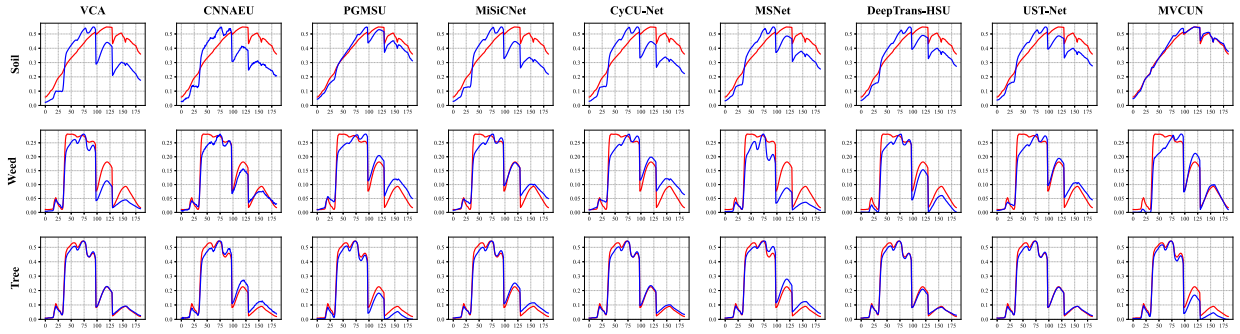


Fig. 16. Visual comparison of extracted endmembers using different unmixing methods and the corresponding ground truth endmembers in the Orchard dataset: blue represents extracted endmembers, red represents ground truth endmembers.

TABLE X

PERFORMANCE OF DIFFERENT METHODS ON THE ORCHARD DATASET. THE BEST PERFORMANCES ARE SHOWN IN RED

Method	SAD(avg)	RMSE(overall)
VCA-FCLS	10.12	26.93
CNNAEU	9.76	31.62
PGMSU	6.56	19.81
MiSiCNet	9.42	24.10
CyCU-Net	9.95	27.83
MSNet	10.06	20.93
DeepTrans-HSU	6.34	13.74
UST-Net	7.05	16.62
MVCUN	**5.85**	**8.23**

extraction. Even when the Metal component constitutes only a small fraction, our method is still able to produce an accurate abundance map, demonstrating the robustness of MVCUN against nonuniform data distributions. Overall, the endmembers and abundance maps produced by our method are highly consistent with the ground truth, further demonstrating its strong unmixing performance in complex scenarios.

F. Experiment With Cuprite Dataset

To more comprehensively evaluate the proposed method, we conducted experiments on the more complex Cuprite dataset [see Fig. 14(a)]. This dataset was acquired by the AVIRIS sensor and consists of 250×191 pixels with 224 spectral bands covering the wavelength range from 400 to 2500 nm. After standard preprocessing, 36 bands affected by atmospheric absorption and noise were removed, resulting in 188 usable bands. The Cuprite region contains a variety of complex minerals. We selected five dominant minerals for unmixing: "Alunite," "Andradite," "Kaolin/Smect," "Sphene," and "Chalcedony." The estimated abundance maps are visualized in Fig. 14(b). The results show that the abundance estimation obtained by our method shows strong spatial consistency with the true-color image, demonstrating the effectiveness of the proposed approach under realistic conditions.

G. Experiment With Orchard Dataset

To validate the effectiveness of the proposed method in nonlinear mixing scenarios, experiments were conducted on the physically-based rendered Orchard dataset [50], [51]. This dataset employs ray tracing technology to fully simulate light transport pathways, inherently incorporating nonlinear mixing effects while providing pixel-level ground truth. The dataset

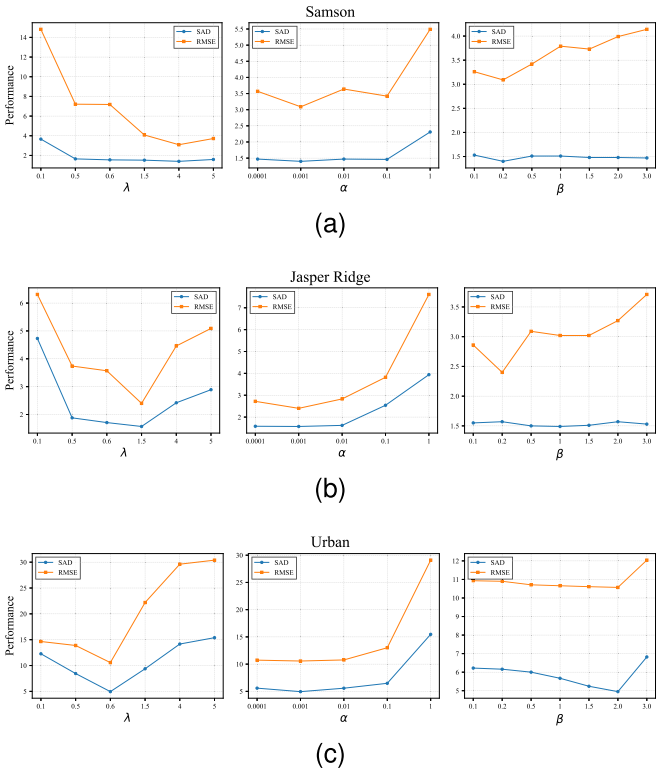


Fig. 17. Effect of varying regularization coefficients on SAD and RMSE for real datasets. (a) Samson dataset. (b) Jasper Ridge dataset. (c) Urban dataset.

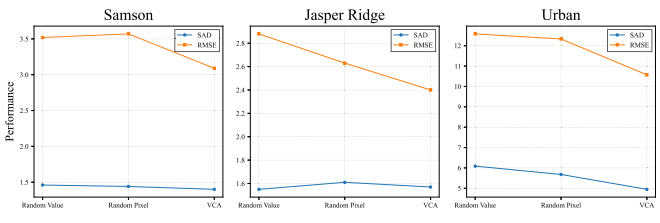


Fig. 18. Comparison of endmember initialization strategies on real datasets.

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TABLE XI
QUANTITATIVE ABLATION STUDY RESULTS. THE BEST PERFORMANCES ARE SHOWN IN RED

	SAD (avg)					RMSE (overall)				
dataset	$\alpha = 0$	$\lambda = 0$	$\beta = 0$	w/o MSGCA	MVCUN	$\alpha = 0$	$\lambda = 0$	$\beta = 0$	w/o MSGCA	MVCUN
Simulated (20 dB)	1.99	2.55	1.73	1.77	**1.71**	2.53	2.51	2.65	2.48	**2.02**
Samson	1.47	11.77	1.44	1.53	**1.40**	3.61	21.19	3.79	3.90	**3.09**
JasperRidge	1.68	13.16	1.64	1.60	**1.57**	2.80	13.62	2.93	3.19	**2.40**
Urban	7.32	15.32	6.41	6.86	**4.95**	12.27	21.65	12.61	13.09	**10.57**

comprises 20 × 20 pixels with a spatial resolution of 2 m, containing three endmembers: Tree, Soil, and Weed [52].

The experimental results are summarized in Table X, with visualizations presented in Figs. 15 and 16. For endmember estimation, the proposed method demonstrates superior overall performance, with its estimation of the soil endmember showing the closest agreement to the reference values. In terms of abundance estimation, our method also achieves optimal performance, accurately capturing the spatial distribution of the actual land cover types. These results confirm that the proposed method can deliver satisfactory outcomes even when dealing with data containing nonlinear mixing effects.

H. Parameter Sensitivity Analysis

To evaluate the impact of hyperparameters in the loss function on the unmixing performance, we conducted a parameter sensitivity analysis on three real datasets. Using a one-at-a-time variation approach—where only one parameter was altered while the others were held fixed—we examined the effects of the TV regularization coefficient λ, the minimum volume constraint coefficient α, and the abundance sparsity regularization coefficient β. The specific results are shown in Fig. 17. Overall, the parameter λ has the most significant influence on model performance and requires careful calibration based on the characteristics of the dataset. The parameter α performs better when assigned smaller values, while β exhibits strong robustness across a wide range of values. The optimal hyperparameter configurations for each dataset are summarized in Table I.

I. Initialization Sensitivity Analysis

To validate the robustness of the method to endmember initialization, we compared three initialization strategies on three real datasets: 1) initialization using random values; 2) randomly selecting pixels from the original image as initial endmembers; and 3) initialization using endmembers extracted by VCA. The results, as shown in Fig. 18, indicate that the choice of initialization method has a certain impact on the unmixing performance. Overall, the VCA initialization method performs the best, demonstrating the importance of accurate endmember initialization for unmixing performance. Both random pixel and random value initialization yield slightly worse unmixing results compared to VCA, with minimal difference between the two. Therefore, it is recommended to prioritize VCA initialization in practical applications, especially when dealing with complex data, to ensure optimal and stable unmixing performance. Furthermore, the proposed method achieves satisfactory results with all three initialization strategies, indicating that it possesses a certain degree of robustness to the choice of initialization approach.

J. Ablation Study

To validate the effectiveness of the proposed constraint terms and the MSGCA module, we conducted ablation studies on real datasets as well as on the simulated dataset with a noise level of 20 dB. Specifically, we analyze each constraint's effect by disabling α, λ, and β individually (set to 0) while keeping other parameters constant. In addition, we tested a model without the MSGCA module to further analyze its contribution to endmember extraction and abundance estimation.

The experimental results are presented in Table XI. Overall, the endmember TV constraint (λ) has the greatest impact, especially in the real datasets; removing this constraint leads to a significant drop in both endmember extraction and abundance estimation performance, indicating its dominant role in the loss function. Moreover, the simplex volume upper-bound constraint (α) also affects the accuracy of endmember extraction, as its removal increases the SAD metrics in certain datasets.

On the other hand, the abundance sparsity constraint (β) and the MSGCA module mainly influence the accuracy of abundance estimation. According to the RMSE metrics, removing either the abundance sparsity constraint or the MSGCA module leads to an increase in RMSE across all datasets. In particular, for the Urban dataset, the RMSE rises from 10.57 to 13.09 when the MSGCA module is removed, and to 12.61 when the abundance sparsity constraint is removed, further demonstrating the importance of these components in complex scenarios.

V. CONCLUSION

In this article, we propose a new blind unmixing autoencoder network named MVCUN. Our network incorporates a minimum volume constraint by including an endmember TV term and an upper bound constraint on the endmember simplex volume in the loss function, thereby enhancing the endmember extraction capability under nonpure pixel scenarios and improving robustness to nonuniform data distributions. Meanwhile, our network also adopts an MSGCA module that extracts local channel information via multiscale cross-channel convolutions and combines it with global channel information obtained through fully connected layers, effectively leveraging the spectral features of hyperspectral images

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to generate superior abundance maps. The experimental results demonstrate that our MVCUN achieves the best performance across all considered methods on both simulated and real datasets. However, our network currently involves numerous hyperparameters, and the optimal values for the constraint coefficients vary across different datasets, posing challenges for hyperparameter tuning. Therefore, one important direction for future work is to optimize the hyperparameter settings.

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